

NFL Predictive Win Model

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Background

- The National Football League (NFL) is the most popular professional team sports league in the United States.
- Similar to other American team sports, such as baseball and basketball, there has been a significant increase in the use of data driven analytics in recent decades to inform team decision making.
- In any professional sports league, there is a strong emphasis on identifying a “**winning formula**,” making the ability to predict team wins highly valuable.
- The purpose of this study is to develop a **predictive model to forecast NFL team wins** based on a set of key predictors.



Background

- Similar predictive models for winning outcomes in the NFL have been developed in prior research.
- Notably:
 - *Advancing NFL Win Prediction: From Pythagorean Formulas to Machine Learning Algorithms* (Weirich et al., 2025)
 - *A Predictive Analytics Model for Forecasting Outcomes in the National Football League Games Using Decision Tree and Logistic Regression* (Gifford & Bayrak, 2023)
- However, none of these studies develop a predictive model that explicitly accounts for the **longitudinal nature of team performance** by modeling the **count outcome of wins over time**.

Goal

- A predictive model was developed to estimate NFL team performance, measured by regular season wins, using historical longitudinal data.
 - The analysis uses data from the 2003–2023 seasons, as the 2002 season was the first year following league expansion to 32 teams and is excluded to ensure structural consistency.
- The model trained on the 2003–2023 seasons is applied to the 2024 season as a testing dataset to evaluate its predictive accuracy.

Data

- The NFL consists of 32 teams.
- **Each NFL team is treated as a cluster** in the analysis.
- Within each cluster, teams contribute one observation per season, resulting in repeated longitudinal measurements over time.



Variables chosen

- The **outcome variable** is the **total number of regular season wins** for each team in a given season.
- Predictor variables were selected based on data availability, subject matter expertise, and prior literature, including the studies discussed in the background section.
- **Defensive rating (Defense)**: Team defensive performance categorized into four equally sized groups, where
 - Category 1 represents teams ranked among the top 8 defenses, and
 - Category 4 represents teams ranked among the bottom 8 defenses.
- **Offensive rating (Offense)**: Team offensive performance categorized into four equally sized groups, where
 - Category 1 represents teams ranked among the top 8 offenses, and
 - Category 4 represents teams ranked among the bottom 8 offenses.

Variables chosen

- **Made playoffs previous year (Playoffs):** An indicator variable denoting whether a team qualified for the playoffs in the previous season.
- **Average age (Age):** The average age of the team's roster in the given season.
- **Coaching tenure (Coaching):** The number of seasons the team's current head coach has been with the team at the start of the given season.

Descriptive statistics

Seasons	Offense; Frequency				Defense; Frequency				Playoffs; Frequency	Age; Mean (SD)	Coaching; Mean (SD)
	1	2	3	4	1	2	3	4			
2003	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.62)	1.84 (0.37)
2004	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.58)	2.41 (0.84)
2005	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.58)	3.16 (1.08)
2006	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.51)	3.06 (1.70)
2007	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.49)	3.34 (2.01)
2008	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	27.0 (0.54)	3.94 (2.20)
2009	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.9 (0.52)	3.44 (2.50)
2010	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.8 (0.57)	4.19 (2.66)
2011	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.7 (0.49)	4.11 (2.87)
2012	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.6 (0.46)	4.33 (3.16)
2013	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.4 (0.51)	4.09 (3.17)
2014	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.4 (0.42)	4.27 (3.48)
2015	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.5 (0.49)	4.38 (3.85)
2016	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.1 (0.41)	4.59 (3.93)
2017	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.46)	5.03 (4.28)
2018	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.41)	5.23 (4.75)
2019	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.0 (0.42)	4.83 (4.52)
2020	25%	25%	25%	25%	25%	25%	25%	25%	37.5%	26.3 (0.47)	4.97 (4.63)
2021	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.3 (0.54)	5.00 (5.04)
2022	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.4 (0.45)	4.59 (5.04)
2023	25%	25%	25%	25%	25%	25%	25%	25%	43.8%	26.5 (0.43)	5.14 (5.27)

SD = Standard Deviation
All statistics calculated on n = 32 observations (32 NFL teams)



Model

- A population averaged marginal count model was fitted using **generalized estimating equations (GEE)** with a Poisson distribution to account for repeated observations within NFL teams across seasons.
- For this model, the following assumptions were assessed:
 - **Dispersion:** Evaluation of overdispersion or under dispersion relative to the Poisson mean–variance assumption.
 - **Outcome type:** The response variable consists of non negative integer counts.
 - **Linearity:** A linear relationship between the predictors and the log rate of the outcome was assumed.

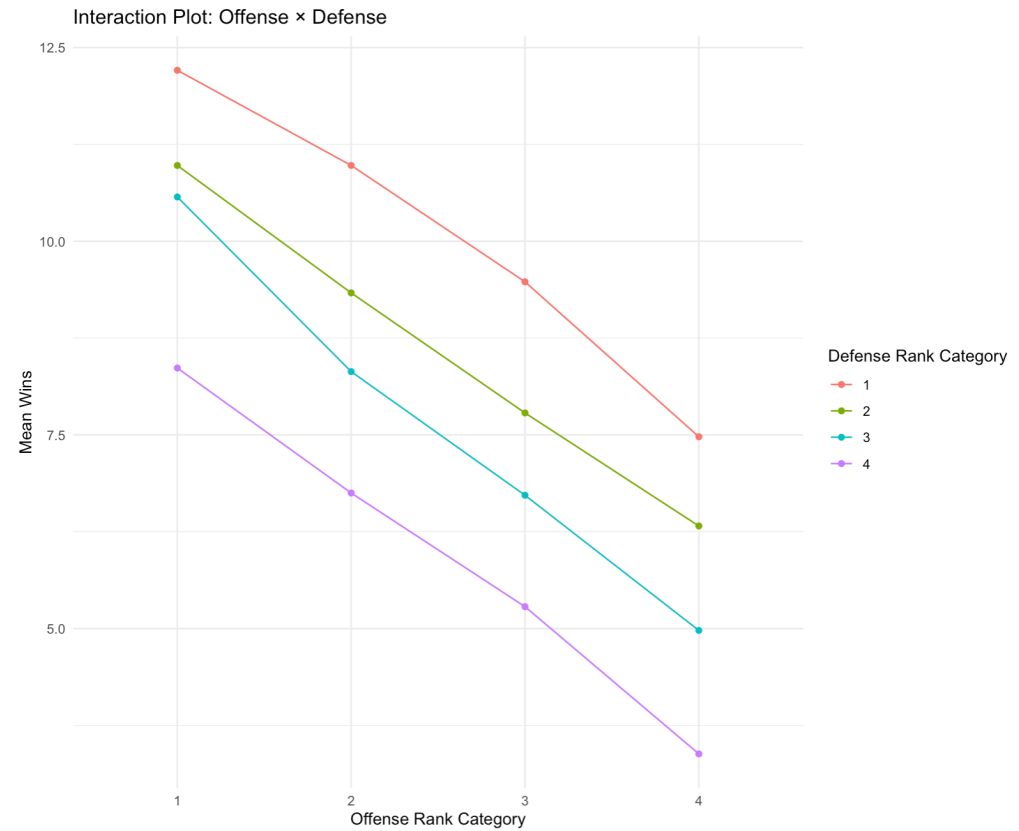
Season	Mean of wins	Variance of wins	Games played
2003	8	9.42	32
2004	8	9.48	32
2005	8	11.48	32
2006	8	8.39	32
2007	8	11.03	32
2008	7.97	11.06	32
2009	8	10.39	32
2010	8	8.97	32
2011	8	10.71	32
2012	7.97	9.52	32
2013	7.97	9.71	32
2014	7.97	10.03	32
2015	8	9.29	32
2016	7.94	10.25	32
2017	8	10.26	32
2018	7.94	8.38	32
2019	7.97	10.29	32
2020	7.97	12.22	32
2021	8.47	8.45	32
2022	8.41	9.6	32
2023	8.5	7.55	32

Model Assumption

- Overdispersion/Underdispersion

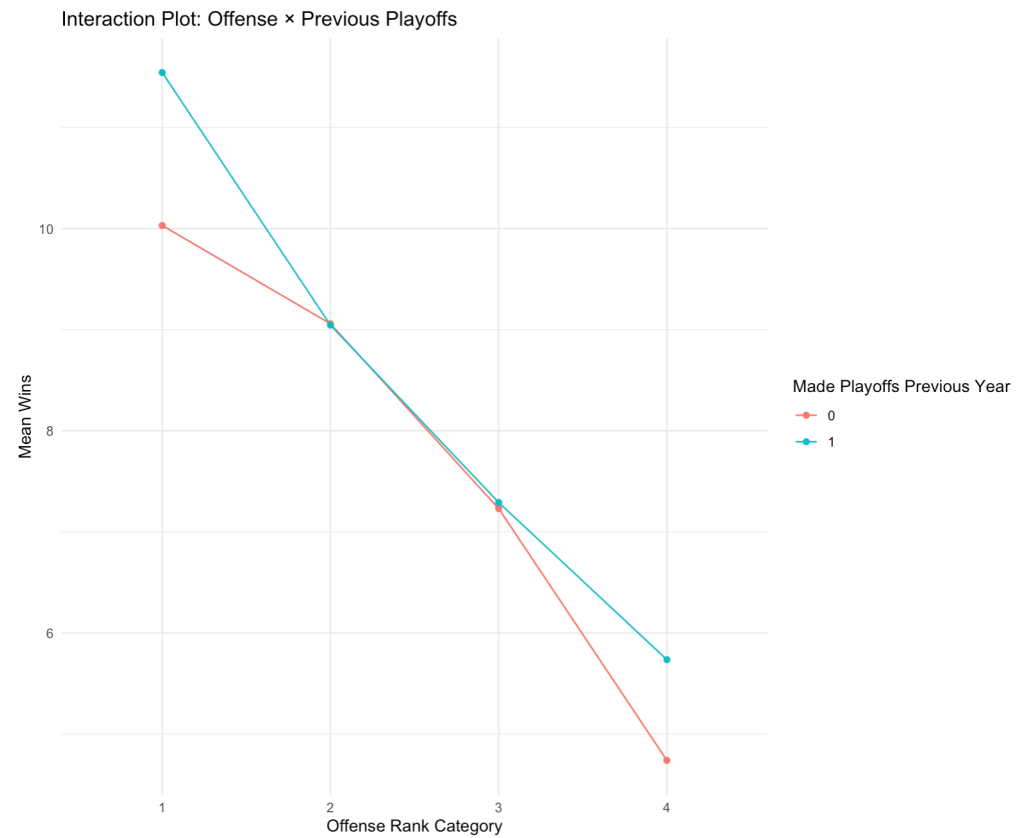
Potential Interactions in Model

OFFENSE VS DEFENSE INTERACTION:



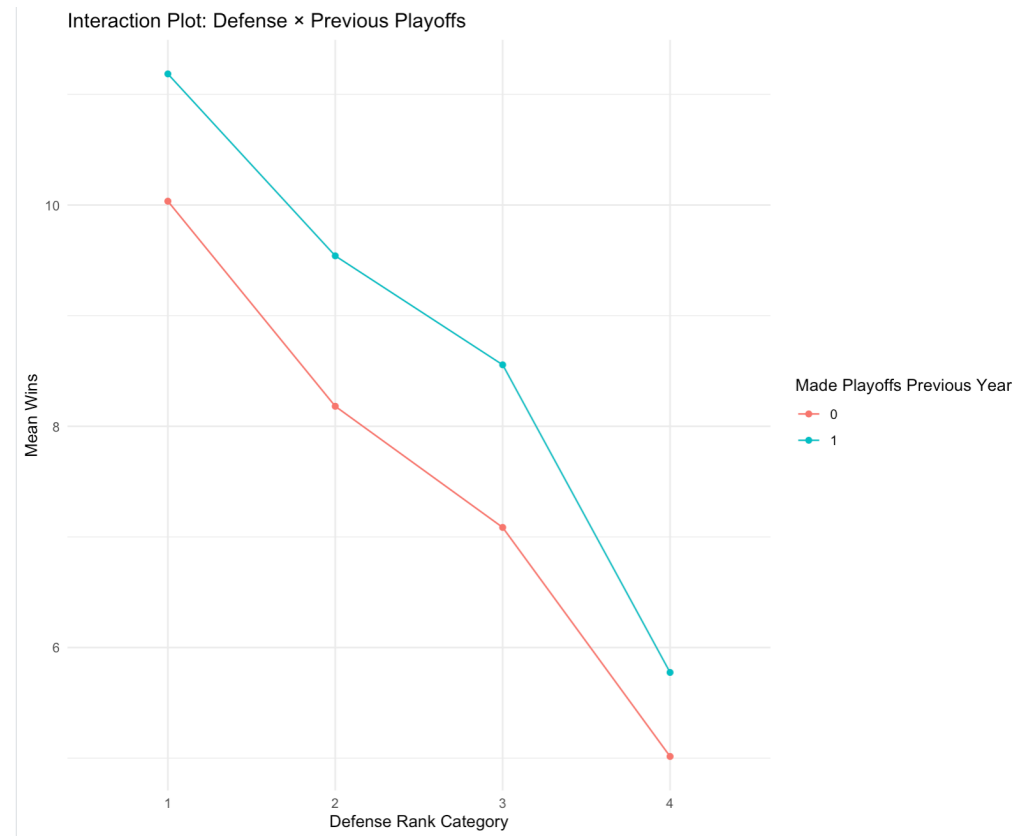
Potential Interactions in Model

OFFENSE VS
PLAYOFFS:



Potential Interactions in Model

DEFENSE VS
PLAYOFFS:



Potential Interactions in Model

- Continuous vs Categorical:
 - To assess potential interaction effects involving **continuous predictors** (coaching and age), separate linear regressions of wins on each continuous variable were fitted **within levels of categorical predictors**.
 - The estimated slopes and standard errors were then compared across groups; **substantial differences in slopes** would suggest effect modification (interaction), while **similar slopes** indicate approximately additive effects.

Potential Interactions in Model

- Age vs Defense:
 - This interaction was considered because the estimated effect of average age on wins varied in both **magnitude and direction** across defensive categories, ranging from **−0.32 to 0.66 wins per additional year of age**.
- Age vs Playoffs:
 - The estimated effect of age differed notably by previous year playoff status, with teams that missed the playoffs showing a much larger positive age effect (**0.79 wins per year**) compared to teams that made the playoffs (**0.12 wins per year**).

Potential Interactions in Model

- Other interaction combinations were not included because subgroup specific slope estimates showed largely parallel relationships, with small differences between minimum and maximum effects indicating no meaningful potential interaction.
- The final potential interactions explored in the model are:
 - Offense vs Playoffs
 - Age vs Defense
 - Age vs Playoffs
 - Age vs Coaching

Comparison within Models

- For model selection, AIC and BIC cannot be used because they rely on the log likelihood, which is not defined for GEE models.
- Instead, the **Quasi likelihood under the Independence model Criterion (QIC)** was used as the appropriate model selection criterion for GEE.
- QIC values cannot be directly compared between **Poisson/Quasi Poisson** and **Negative Binomial** models because they are not computed under the same quasi likelihood framework.
- Therefore, QIC was used to compare models **within each model type**, and backward selection was applied to identify the best performing model for each distribution.

Model used for backwards selection

- Main effects:
 - Defense, Offense, Playoffs, Age, Coaching
- Interactions:
 - Offense vs Playoffs
 - Age vs Defense
 - Age vs Playoffs
 - Age vs Coaching

Model used for backwards selection

- Y_{it} = number of wins for team i in season t
- G_{it} = games played (offset)
- $Y_{it} \sim \text{Count}(\mu_{it})$

$$\begin{aligned} \log(\mu_{it}) = & \log(G_{it}) + \beta_0 + \beta_1 \text{Offense}_{it} + \beta_2 \text{Defense}_{it} + \beta_3 \text{Playoffs}_{i,t-1} + \beta_4 \text{Age}_{it} + \beta_5 \text{Coaching}_{it} \\ & + \beta_6 (\text{Offense}_{it} \times \text{Playoffs}_{i,t-1}) + \beta_7 (\text{Age}_{it} \times \text{Defense}_{it}) + \beta_8 (\text{Age}_{it} \times \text{Playoffs}_{i,t-1}) + \beta_9 (\text{Age}_{it} \times \text{Coaching}_{it}) \end{aligned}$$

Model used for backwards selection

- **Log(offset):**

The term `offset(log(games played))` was included to adjust for differences in the number of regular season games played across seasons, allowing the model to estimate a **rate of wins per game** rather than raw win counts and ensuring comparability across seasons.

- **Correlation:**

A working correlation structure was specified to account for within team correlation arising from repeated observations across seasons. An unstructured correlation matrix was considered but not used due to convergence issues. The remaining correlation structures (**independence, exchangeable, and AR(1)**) were evaluated during backward model selection.

Model used for backwards selection

- **ID = team:**
Each NFL team was specified as a clustering unit to appropriately model repeated measurements for the same team within the GEE framework.
- **Waves = season:**
The season variable was used as the time index to preserve the temporal ordering of observations within each team.
- **Outcome:**
The response variable is the total number of regular season wins per team per season.
- **Data used:**
The model was fitted using the training dataset spanning the 2003–2023 seasons, which includes one observation per team per season under the current 32 team NFL structure.

Model chosen for Poisson

- Main effects:
 - Defense, Offense, Playoffs, Age, Coaching
- Interactions:
 - Offense vs Playoffs
 - Age vs Defense
 - Age vs Playoffs
 - Age vs Coaching



Model chosen for Quasi Poisson

- Main effects:
 - Defense, Offense, Playoffs, Age, Coaching
- Interaction:
 - Age vs Coaching

Model chosen for Negative Binomial

- Main effects:
 - Defense, Offense, Playoffs, Age, Coaching
- Interaction:
 - Offense vs Playoffs
 - Age vs Coaching

Comparison between each top model

- Root Mean Square Error (RMSE) was used to evaluate the predictive performance of each model based on the Training Data.
- Results:
 - Poisson GEE: RMSE = 1.62
 - Quasi Poisson GEE: RMSE = 1.63
 - Negative Binomial GEE: RMSE = 4.15
- The Poisson and Quasi Poisson GEE models exhibit very similar predictive performance
- The substantially higher RMSE for the Negative Binomial GEE model indicates poorer predictive accuracy on the training data



Comparison between Poisson and Quasi Poisson Model

- RMSE was again used to assess predictive performance using the Testing Data (2024 season).
- Results:
 - Poisson GEE: RMSE = 1.65
 - Quasi Poisson GEE: RMSE = 1.59
- The Quasi Poisson GEE model was retained as the final predictive model, as it achieved the lowest RMSE on the testing data, indicating superior out of sample predictive performance.

Team	Actual Wins	Predicted Wins (Poisson GEE)	Predicted Wins (Quasi-Poisson GEE)
ARI	8	9.98	9.77
ATL	8	9.11	8.99
BAL	12	13.45	13.43
BUF	13	11.64	11.72
CAR	5	5.78	5.61
CHI	5	5.93	6.01
CIN	9	7.55	7.79
CLE	3	4.47	4.13
DAL	7	5.46	5.63
DEN	10	11.27	11.25
DET	15	11.8	11.87
GB	11	13.61	13.7
HOU	10	8.1	8.51
IND	8	8.84	8.78
JAX	4	3.9	4.02
KC	15	11.65	11.8
LA	10	8.3	8.44
LAC	11	11.19	11.18
LV	4	3.76	3.93
MIA	8	8.04	8.51
MIN	14	11.62	11.49
NE	4	5.09	5.23
NO	5	7.6	7.42
NYG	3	5.22	5.35
NYJ	5	7.57	7.39
PHI	14	13.6	13.6
PIT	10	8.77	9.03
SEA	10	8.39	8.1
SF	6	6.69	6.58
TB	10	10.51	10.48
TEN	3	3.96	4.05
WAS	12	10.79	10.79

Comparison between Poisson and Quasi Poisson Model

TABLE OF PREDICTIONS
ON TESTING DATA
(2024 SEASON):

Final Model

- The Quasi Poisson GEE model was selected as the final model among the three candidates.
 - Y_{it} = number of wins for team i in season t
 - G_{it} = games played (offset)
 - $Y_{it} \sim \text{Count}(\mu_{it})$
- $$\log(\mu_{it}) = \log(G_{it}) + \beta_0 + \beta_1 \text{Offense}_{it} + \beta_2 \text{Defense}_{it} + \beta_3 \text{Playoffs}_{i,t-1} + \beta_4 \text{Age}_{it} + \beta_5 \text{Coaching}_{it} + \beta_6 (\text{Age}_{it} \times \text{Coaching}_{it})$$

Final Model

Term	Estimate	Std. Error	Significance
Intercept	-1.8592	0.5953	**
Offensive Rating (Category 2 vs 1)	-0.1616	0.0186	***
Offensive Rating (Category 3 vs 1)	-0.3428	0.0225	***
Offensive Rating (Category 4 vs 1)	-0.6536	0.0296	***
Defensive Rating (Category 2 vs 1)	-0.1424	0.0187	***
Defensive Rating (Category 3 vs 1)	-0.2544	0.0208	***
Defensive Rating (Category 4 vs 1)	-0.5408	0.0280	***
Made Playoffs Previous Season (Yes vs No)	0.0203	0.0167	—
Average Team Age (years)	0.0612	0.0224	**
Coaching Tenure (years)	0.2923	0.1091	**
Average Age × Coaching Tenure	-0.0111	0.0041	**
Significance: *** p < 0.001 ** p < 0.01 * p < 0.05			

Conclusion

- This study developed a **longitudinal, population averaged model** for predicting NFL regular season wins, accounting for repeated observations within teams across seasons using GEE.
- Among competing models, the **Quasi Poisson GEE** achieved the best overall predictive performance and robustness, demonstrating the value of **count based longitudinal modeling** for forecasting NFL team performance.
- The model performed well on the 2024 season, with an **RMSE of 1.59**, indicating that predicted win totals were, on average, within **approximately 1–2 wins** of actual outcomes, reflecting **reasonable and practically meaningful predictive accuracy**.

References

- Weirich C, Kim JW, Yoon Y, Jeong S. Advancing NFL win prediction: from Pythagorean formulas to machine learning algorithms. *Front Sports Act Living*. 2025 Sep 12;7:1638446. doi: 10.3389/fspor.2025.1638446. PMID: 41018816; PMCID: PMC12463883.
- Matt Gifford, Tuncay Bayrak, A predictive analytics model for forecasting outcomes in the National Football League games using decision tree and logistic regression, *Decision Analytics Journal*, Volume 8, 2023,100296, ISSN 2772-6622, <https://doi.org/10.1016/j.dajour.2023.100296>.